
A Neural Verifier for Structured Table Extraction - Milestone Report

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1 Introduction

Large Language Models (LLMs) and vision-language models have achieved remarkable performance in document understanding, yet remain unstable in structured data extraction. For example, while models such as TATR-v1.0 achieve near-perfect structural recognition on PubTables-1M (GriTS ≈ 0.985), their semantic content accuracy remains imperfect ($\text{Acc}_{\text{Con}} \approx 0.82$). Even state-of-the-art multimodal systems such as GPT-4o reach less than 90% Grid Table Similarity (GriTS F1) (2), leaving a nontrivial reliability gap. This suggests that while table geometry is largely solved, semantic correctness and consistency across extracted cells still present challenges. This instability limits the use of LLMs in high-stakes workflows such as legal, financial, or medical document analysis, where structured accuracy and auditability are essential. The goal of this project is to train a neural **Verifier model** that automatically evaluates the quality of structured table extraction from PDFs. Instead of manually comparing extracted tables to ground truth, the model will learn to predict extraction quality—quantifying semantic correctness, schema alignment, and numerical consistency. This Verifier can serve as a foundation for improving reliability, benchmarking models, or guiding future optimization.

2 Dataset Curation

We created our dataset using by SciTSR dataset as the upstream. It has 12000 training data points and 3000 validation data points from science journals. Each data point is a table with the annotated ground truth JSON structure. We use Table Transformer and EasyOCR to generate the JSON files from the same table and calculate a similarity score as described below:

```
0.50 * f1           # Cell detection (primary metric)
0.25 * row_accuracy # Grid structure
0.25 * col_accuracy # Grid structure
```

A sample data point looks like below:

```
"id": "9608007v1.5",
"ground_truth_file": "9608007v1.5.json",
"generated_file": "9608007v1.5.json",
"similarity_score": 0.5641891891891893
```

The ground_truth file and generated_file contain the JSON structure. The similarity_score is the label we will train the model to learn. Then we convert the 15000 data points into the data parquet format and upload to HuggingFace.

The link to our curated dataset: Table extraction evaluation dataset

With this dataset, we defined this table extraction evaluation as a supervised learning problem; the model will learn to predict the similarity score, which serves as an evaluation model for unseen JSON table structure files.

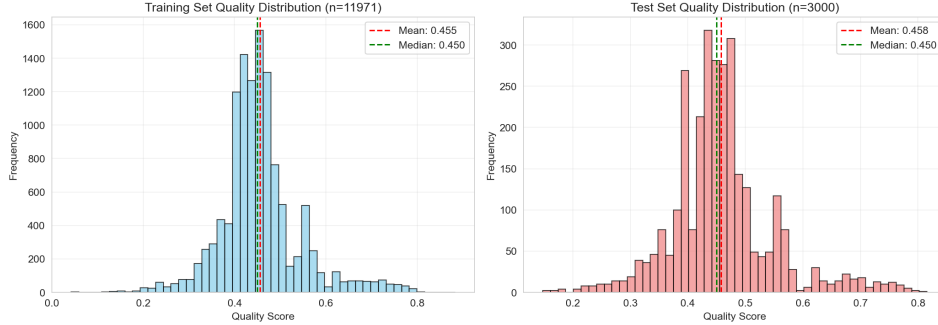


Figure 1: The training and test set quality distribution

3 Model Details

We are currently using a 2-layer MLP with the Neural Network Architecture as follows:

Input $\rightarrow$ FC1 (768) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout (0.05) $\rightarrow$ FC2 (384) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout (0.2) $\rightarrow$ Output (1) $\rightarrow$ Sigmoid

We use the semantic Embedding model from Sentence-transformers sentence-transformers/all-mpnet-base-v2 with 768 dimension size.

Training Configuration

- Optimizer: Adam with weight decay (1×10^{-5})
- Loss Function: Mean Squared Error (MSE)
- Learning Rate: 0.0002 with ReduceLROnPlateau scheduler
- Batch Size: 64
- Epochs: 100 (with early stopping patience of 15)
- Gradient Clipping: Max norm 1.0
- Regularization:
 - Input dropout: 0.05
 - Hidden dropout: 0.1
 - Batch normalization on all hidden layers
 - L2 weight decay: 1×10^{-5}

Evaluation Metrics:

- **Mean Squared Error (MSE):** Primary training loss
- **Mean Absolute Error (MAE):** Average prediction error
- **Root Mean Squared Error (RMSE):** Standard deviation of residuals
- **R² Score:** Coefficient of determination
- **Accuracy within 5%:** Percentage of predictions within 5% of ground truth
- **Accuracy within 10%:** Percentage of predictions within 10% of ground truth

4 Intermediate Results

4.1 Baseline

Our baseline MLP model achieved the following performance on the validation set after training:

Metric	Value
MSE	0.007
MAE	0.058
RMSE	0.084
R ² Score	0.077
Accuracy (within 1%)	16.1%
Accuracy (within 5%)	59.4%
Accuracy (within 10%)	82.7%

Table 1: Baseline model performance on validation set

4.2 Training curves (loss vs epoch)



Figure 2: Training History

5 Next Steps

To further enhance the performance and robustness of our Verifier model, we outline the following next steps organized by technical domain.

5.1 1. Model Architecture and Training

- **Embedding Exploration:** Experiment with structure-aware embeddings such as Tree-LSTMs, or transformer-based encoders that can model the hierarchical relationships within JSON table structures.
- **Loss Function Refinement:** Introduce correlation-based or ranking-based loss to better align predicted and true similarity scores.

5.2 2. Data and Evaluation

- **Distribution Balancing:** Apply weighted sampling or oversampling for low-frequency quality scores to address class imbalance.
- **Metric Expansion:** Evaluate with correlation metrics (Pearson, Spearman), calibration error, and ranking accuracy to provide a more comprehensive understanding of model reliability.

6 Code

The complete source code, including data processing scripts, model implementations, and training pipelines, is available at: Source code of the evaluation model

References

- [1] Chi et al., *Complicated Table Structure Recognition*, arXiv, 2019.
- [2] Silva et al., *Benchmarking Table Extraction: A Comparative Study of Multimodal Models*, Universidade NOVA de Lisboa, 2024. https://run.unl.pt/bitstream/10362/185720/1/Benchmarking_table_extraction.pdf
- [3] Xu et al., *LayoutLMv3: Pretraining for Document AI*, arXiv, 2023.
- [4] Kim et al., *Donut: OCR-Free Document Understanding Transformer*, ECCV, 2022.

This is appendix